

Quantum Born Machines for Local Life-and-Death Inference in the Game of Go

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April 2026

Abstract

We introduce a quantum-inspired framework for evaluating local life-and-death problems in the game of Go. Instead of relying on classical search or Monte Carlo simulation, we model tactical configurations as quantum superpositions of possible responses. A Born Machine generates a probability amplitude distribution over candidate moves, while life-and-death status is computed as the expectation value of a classical observable. This approach provides a compact representation of local tactical inference and suggests a new perspective on reading in Go as a form of quantum probabilistic reasoning.

1 Introduction

Life-and-death evaluation in Go is a paradigmatic example of combinatorial reasoning: highly local, deeply structured, and sensitive to small tactical variations. Traditional engines rely on tree search, Monte Carlo simulation, or deep neural networks trained on large datasets. However, local tactical reading often resembles evaluating multiple hypothetical futures simultaneously, a structure reminiscent of quantum superposition.

We propose a model where:

1. the board is classical,
2. the moves of the attacking player are represented as a quantum state,
3. the life status of a group is an observable evaluated over the superposition.

This yields a compact, differentiable, and interference-capable representation of local Go reasoning.

2 Classical Board Representation

Let the board be a grid $B \in \{0, 1, 2\}^{N \times N}$ with:

$$0 = \text{empty}, \quad 1 = \text{black}, \quad 2 = \text{white}.$$

A group G is a connected component under 4-neighborhood adjacency. Liberties are defined as:

$$L(G) = \{p \in B : p \text{ adjacent to } G \text{ and } B[p] = 0\}.$$

Life status is:

$$\text{life}(G) = \begin{cases} 1 & \text{if } |L(G)| > 0, \\ 0 & \text{otherwise.} \end{cases}$$

3 Quantum Move Superposition (Born Machine)

Let the attacker have M candidate moves $\{m_1, \dots, m_M\}$. We encode them in a quantum register of $k = \lceil \log_2 M \rceil$ qubits:

$$|\Psi(\boldsymbol{\theta})\rangle = \sum_{i=1}^M \alpha_i(\boldsymbol{\theta}) |i\rangle,$$

where amplitudes are generated by a parameterized quantum circuit:

$$\alpha(\boldsymbol{\theta}) = U(\boldsymbol{\theta})|0\rangle^{\otimes k}.$$

A typical ansatz is:

$$U(\boldsymbol{\theta}) = \prod_{j=1}^k R_y(\theta_j) \prod_{j=1}^{k-1} \text{CX}_{j,j+1}.$$

4 Life Observable as Expected Value

For each move m_i , we compute the resulting board B_i and life status:

$$\ell_i = \text{life}(G \mid m_i).$$

The quantum expected value is:

$$\mathbb{E}[\text{life}] = \sum_{i=1}^M |\alpha_i(\boldsymbol{\theta})|^2 \ell_i.$$

This is the quantum life-and-death evaluation.

5 Training Objective

We optimize $\boldsymbol{\theta}$ to minimize the life expectation:

$$\boldsymbol{\theta} = \arg \min_{\boldsymbol{\theta}} \mathbb{E}[\text{life}].$$

Gradient-free methods (SPSA, finite differences) are effective due to the small parameter count.

6 Compression for 9×9 Boards

A full quantum encoding of a 9×9 board would require:

$$81 \text{ points} \times 2 \text{ qubits} = 162 \text{ qubits},$$

which is infeasible.

We instead compress by:

- keeping the board classical,
- keeping only the move register quantum,
- evaluating life/morte classically per branch.

This reduces the quantum footprint to:

$$k = \lceil \log_2 M \rceil \leq 6 \text{ qubits}.$$

7 Results (Preliminary)

Experiments on standard shapes (bulky five, false eye, snapback) show:

- the Born Machine converges to kill moves,
- the life expectation approaches 0,
- the model generalizes across local patterns.

8 Conclusion

We present a quantum-inspired framework for local tactical inference in Go. By treating move selection as a quantum superposition and life/morte as an observable, we obtain a compact and expressive model that avoids explicit search. This suggests new directions for hybrid quantum-classical reasoning systems.

References

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